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Questions

1. The three fully-connected layers have:

FC 1: $25088 \times 4096 + 4096 \approx 102.8\ M$

FC 2: $4096 \times 4096 + 4096 \approx 16.8\ M$

FC 3: $4096 \times 1000 + 1000 \approx 4.1\ M$

adding up to about 123.6 million parameters.

Layer by layer in VGG 16:

Conv 1_1 (3→64): $(3 \times 3 \times 3 + 1) \times 64 = 1792$

Conv 1_2 (64→64): $(3 \times 3 \times 64 + 1) \times 64 = 36928$

Conv 2_1 (64→128): $(3 \times 3 \times 64 + 1) \times 128 = 73856$

Conv 2_2 (128→128): $(3 \times 3 \times 128 + 1) \times 128 = 147584$

Conv 3_1 (128→256): $(3 \times 3 \times 256 + 1) \times 256 = 295168$

Conv 3_2 (256→256): $(3 \times 3 \times 256 + 1) \times 256 = 590080$

Conv 4_1 (256→512): 590080

Conv 4_2 (512→512): $(3 \times 3 \times 256 + 1) \times 512 = 1180160$

Conv 4_3 (512→512): $(3 \times 3 \times 512 + 1) \times 512 = 2359808$

Conv 5_1, Conv 5_2, Conv 5_3: 2359808

adding up to about 1.47×10^7 parameters

Total = conv + FC $\approx 1.38 \times 10^8$

So, VGG 16 has about 1.38×10^8 parameters in total:

roughly 1.24×10^8 in the fully-connected layers and

1.47×10^7 in the convolutional layers.

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2. The last layer outputs a 1000-dimensional vector.
It corresponds to the score (logits) for the 1000 ImageNet classes.

3. On natural images, VGG16 usually produces resonable top predictions that match the main object in the image. However, when the object is not exactly one of the ImageNet classes, the model tends to choose the closest-looking category based on texture and shape. For example, when I provided an image of a tower, the network predicted "crane", which is visually similar in structure but semantically incorrect. These mistakes show that VGG16 does not truly understand the object but relies on visual resemblance within the 1000 ImageNet categories.

- role of the ImageNet normalization:
ImageNet normalization rescales the pixels and seandardizes each RGB channel so that the input diseribution matches the one used to train VGG16. This makes the pre-trained net work properly and keeps the activations in a seable range.

- Why setting to eval model:
model.eval() puts the network in inferenca mode: it disables dropout and makes batch-normalization layers use their learned running statistics instead of updating them. This ensures deterministic and consistent predictions when we only want to evaluate the model.

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4. The activation maps of the first convolutional layer highlight low-level visual pattern such as edges, color contrasts, and simple textures. Each filter responds to a different type of primitive feature. Some maps activate strongly on vertical or horizontal edges, while others respond to color changes, corners or simple gradients. These early features form the basis for deeper layers to build more complex object representations.

5. Because the 15 Scene dataset is too small to train a deep model like VGG16 from scratch. Training all 138 million parameters would lead to strong overfitting, requires large computational resources and would not achieve good generalization.

6. Pretraining on ImageNet provides strong general visual features such as edges, textures, shapes and object parts. These representations transfer well to new tasks, even with small datasets. Using these pre-trained features from 15 Scene greatly improves performance and prevents overfitting.

7. Feature extraction is limited because the pre-trained network is fixed and cannot adapt to the new task. The extracted features may not be optimal for the 15 Scene dataset. Additionally, high-level information specific to the new classes cannot be learned, and performance is generally lower than fine-tuning when sufficient data is available.

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8. Features extracted from early layers capture low-level information such as edges, textures or color patterns, while features from deeper layers represent more abstract and semantic concepts. Choosing an earlier layer gives more generic but less discriminative features; choosing a deeper layer gives more task-specific and powerful representations, but they may be less transferable. Therefore, the extraction layer controls the trade-off between generality and discriminative power.

9. We can convert each grayscale image into a 3-channel RGB image by duplicating the single channel three times. This produces an image of shape (3, H, W), which is compatible with VGG16 while preserving the original visual information.

10. Yes, it is possible to use the neural network directly, but only if we fine-tune it on the new dataset. The pre-trained VGG16 classifier is specialized for the 1000 ImageNet classes and cannot classify the 15 scene categories without retraining. Fine-tuning the last layer would adapt the model to the new task. However, this requires much more data, computational cost, and careful training to avoid overfitting. Therefore, using an external classifier such as an SVM on top of fixed deep features ~~or~~ is more stable for small datasets like 15 Scene.

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11. Changing the layer where features are exteracted affects how semantic the representation is. Features from Relu 6 (4096 dimensions) are less specialized for ImageNet and contain more general mid-level information. In my experiment, relu 6 has a higher accuracy than relu 7, meaning that these more general features transfer better to the 15 Scene task. Deeper layers like relu 7 are more semantic but also more tied to ImageNet classes, which can reduce transfer performance.

- I tested several alternative pre-trained networks.
  Models such as ResNet-50, DenseNet-121, and EffectiveNet-B0 produced compact and highly semantic deep features (2048, 1024 and 1280 dimensions respectively).
  These architectures include skip connections or dense connections and generally extract richer representations than VGG16.
  Using these networks can improve accuracy and reduce feature size and computation time.

- The parameter C controls the balance between margin size and classification errors in the SVM.
  A larger C fits the training data extensively, while a smaller C increases regularization.

- I replaced the last fully-connected layer of VGG16 with a new 4096 → 15 classifier and trained only this layer. The model reached 96-97% training accuracy and ≈ 88.8% validation accuracy, which shows that the ImageNet features are very discriminative for 15 Scene.

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However, the gap between training and validation accuracy indicates clear overfitting, even though only the last layer was trained. This method works well but does not generalize as cleanly as using fixed deep features with an SVM.

- I applied PCA to reduce the 4096 dimension relu7 features before training the SVM.
  Using 512 dimensions reached 0.87 accuracy, almost the same as the full feature while running much faster.
  Reducing to 256 dimensions gave 0.85 accuracy, showing a small drop due to information loss but even lower computation time.
  Overall, PCA provides a good trade-off: moderate reduction keeps high performance, while aggressive reduction speeds up training but harms accuracy.

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